

Ex no: 01    N – Queen's Problem

Program:

N = 8

```
def print_solution(board):  
    for row in board:  
        print(" ".join("Q" if col else "." for col in row))  
    print("\n")
```

```
def is_safe(board, row, col):
```

```
    for i in range(row):  
        if board[i][col]:  
            return False
```

```
    for i in range(row):  
        j = col - (row - i)  
        if j >= 0 and board[i][j]:  
            return False
```

```
    for i in range(row):  
        j = col + (row - i)  
        if j < N and board[i][j]:  
            return False
```

```
    return True
```

```
def solve_n_queens(board, row):
```

```
    if row == N:
```

```
        print_solution(board)
```

```
        return True
```

```
    found = False
```

```
    for col in range(N):
```

```
        if is_safe(board, row, col):
```

```
            board[row][col] = 1
```

```
            found = solve_n_queens(board, row + 1) or found
```

```
            board[row][col] = 0
```

```
    return found
```

```
board = [[0] * N for _ in range(N)]
```

```
solve_n_queens(board, 0)
```

Output:

A 10x10 grid of dots with the letter 'Q' placed at various positions, illustrating a sparse matrix.